

Dr. Babasaheb Ambedkar Technological University, Lonere
B.Tech. course in Electronics and Telecommunication Engineering
Semester Examination, May - 2019

Subject: Embedded Systems Design (BET803)

Semester: VIII

Time : 3 Hours

Marks: 70

Instructions to students:

- 1) Figures to the right in square bracket indicates full marks.
- 2) If some part or parameter is noticed to be missing, appropriate data may be assumed and should be mentioned clearly.
- 3) Q.1 is compulsory. Attempt any 5 questions from remaining.

Q. 1 Choose correct answer from given alternatives.**[10]**

- I. Which of the following is the characteristic of a RISC system?
 a) Barrel Shifter b) Pipelining c) Variable cycle execution d) Conditional Execution
- II. Which of the following unit does not accept values from barrel shifter.
 a) ALU b) MAC c) Incrementer d) All of these
- III. Register r14 is traditionally used as _____.
 a) Link register b) Stack Pointer c) Accumulator d) Program Counter
- IV. Which of the following section of CPSR register gets only read access in non-privileged mode?
 a) Flags b) Status c) Extension d) Control
- V. Which of the following is non-privileged mode.
 a) Abort b) System c) Undefined d) User
- VI. Which of the following instruction will modify CPSR register?
 a) MOV b) ADD c) CMN d) MVN
- VII. While using PLL, CCO should operate in the range _____.
 a) 1MHz to 50 MHz b) 10MHz to 25MHz c) 1MHz to 30 MHz d) 156MHz to 320MHz
- VIII. PLL generator allows running ARM at ____ speed with ____ frequency oscillator connected?
 a) high, low b) low, high c) high, high d) low, low
- IX. Feed values need to be provided to PLLxFEED are ____ and then ____.
 a) 0xAA, 0x55 b) 0x55, 0xAA c) 0xAA, 0xAA d) 0x55, 0x55
- X. Which of the following is read-only register?
 a) PLLxCON b) PLLxCFG c) PLLxSTAT d) PLLxFEED

Q. 2 Attempt following questions.

- a) Draw and explain 32 bit encoding format for data processing instructions. **[6]**
- b) Find out 32 bit encoding values for following instructions. Assume that first conditional bits are 1111 for all instructions listed below. **[6]**
 - i) ADD r1, r2, r3, LSL #22 (Assume that Opcode bits are 1011)
 - ii) SUBS r4, r2, r1, RRX (Assume that Opcode bits are 1110)
 - iii) ADCS r7, r5, r4, ASR r6 (Assume that Opcode bits are 1010)

Q. 3 Attempt following questions.

- a) State different properties of RISC system and explain how ARM differ from pure RISC definition. **[6]**

- b) Assume that five 32 bit numbers are stored from 0x000FFFF0 memory location. Write an assembly language program to find out 1's complement of each and store result of each complement respectively from 0x000EEEE0. [6]

Q. 4 Attempt following questions.

- a) Briefly explain IOxDIR, IOxSET & IOxCLR registers with suitable examples of each. [6]
b) Explain following instructions with suitable examples of each. [6]
i) BIC ii) UMULL iii) LDR

Q. 5 Attempt following questions.

- a) Explain pseudoinstructions of ARM with suitable examples of each. [6]
b) Write a C function to setup PLL of LPC2148 to generate 40MHz CCLK and 20MHz PCLK [6]

Q. 6 Attempt following questions.

- a) Explain different addressing modes of LDM and STM instructions with suitable examples of each. [6]
b) Explain software interrupt instruction of ARM along with its 32 bit encoding format and SWI handler [6]

Q. 7 Attempt following questions.

- a) Write a C program to generate a square wave with 75% duty cycle on pin P1.16 [6]
b) Interface a 7 segment display to LPC2148. Write a C program to display numbers from 0 to 9 repeatedly. [6]

Believe you can and you are halfway there...

All The Best

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